

# Sentinel AI

## **AI-Powered Behavioral Anomaly Detection for Real-Time Cybersecurity & Security Operations**

*A next-generation UEBA platform that learns normal user and device behavior, detects insider threats and credential-based attacks in real time, and explains every alert in the language of the SOC.*

|                      |                                              |
|----------------------|----------------------------------------------|
| <b>Project Title</b> | <b>Sentinel AI — Behavioral SOC</b>          |
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| <b>Category</b>      | <b>AI / Machine Learning · Cybersecurity</b> |

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## Summary

Modern enterprises generate thousands of authentication and access events every minute across cloud platforms, VPNs, identity providers, and endpoint devices. **Traditional signature- and rule-based defenses are inherently reactive** — they only recognize attacks that have already been catalogued, leaving organizations exposed to insider threats, credential misuse, device spoofing, and zero-day behavior. **Sentinel AI** replaces this static model with **User & Entity Behavior Analytics (UEBA)**: it learns a personalized behavioral baseline for every user and flags deviations from that baseline in real time — catching threats that have no known signature.

At its core is a **hybrid two-stage detection pipeline**. Stage 1 is an unsupervised anomaly scorer that answers “is this event abnormal at all?” and works from day one with no labels — so it can surface novel attack patterns. Stage 2 is a supervised **LightGBM** classifier that answers “which known attack type does it most resemble?”, sorting anomalies into five categories: **Credential Misuse, Brute Force, Lateral Movement, Impossible Travel, and Device Spoofing**.

The platform models behavior through **16 behavioral-intelligence features** spanning temporal, geographical, authentication, device, and resource-access dimensions, and layers on production-grade capabilities: **Explainable AI (SHAP), MITRE ATT&CK enrichment, automated response playbooks, severity-based triage, cold-start handling, and behavioral-drift monitoring**.

A synthetic enterprise dataset of **20,351 authentication events** (including **1,151 labeled attacks**) was generated to validate the approach. The anomaly stage achieved a **ROC-AUC of 0.9855**, 87.1% precision, 76.3% recall, and an 81.3% F1-score, while the supervised classifier reached **100% macro precision, recall, and F1** on held-out attack data. The full solution ships as a **FastAPI** backend with **WebSocket** live-alert streaming and an interactive **SOC dashboard** featuring an AI Security Copilot, an investigation inspector, explainable risk scores, and geospatial threat visualization.

### Results at a Glance

| Anomaly ROC-AUC | Anomaly F1-Score | Classifier Macro-F1 | Events Evaluated |
|-----------------|------------------|---------------------|------------------|
| 0.9855          | 81.3%            | 100%                | 20,351           |

# 1. Proposed Solution

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## 1.1 Problem Summary

Enterprise identity is the new perimeter. Attackers increasingly **log in rather than break in** — abusing valid credentials, moving laterally between hosts, and masquerading as trusted devices. **Signature-based tools and static SIEM rules** cannot see these threats because the individual actions look legitimate; only the **behavioral context** reveals the attack. Security teams therefore need a system that learns what “normal” looks like per user, detects deviations in real time, and — critically — explains each alert clearly enough to cut through alert fatigue and accelerate incident response.

## 1.2 High-Level Solution

Sentinel AI is an AI-powered Behavioral Anomaly Detection platform that continuously learns normal user and device behavior instead of relying on predefined signatures. It ingests authentication and access logs, extracts 16 behavioral features, and builds a personalized baseline for every user. An unsupervised anomaly model assigns a behavioral risk score to each event; a supervised LightGBM classifier then identifies the specific attack type. Every alert is automatically enriched with a severity level, a MITRE ATT&CK technique, and a ranked response playbook, then streamed live to an interactive SOC dashboard backed by an AI Security Copilot.

## 1.3 How It Addresses the Problem

- **Personalized behavioral baselines** — learns a per-user profile of typical hours, locations, devices, resources, and session behavior, enabling detection of previously unseen and evolving threats.
- **Real-time multi-attack detection** — surfaces Credential Misuse, Brute Force, Lateral Movement, Impossible Travel, and Device Spoofing as events stream in.
- **Explainable, analyst-ready alerts** — generates SHAP-based, human-readable risk explanations plus MITRE ATT&CK context and recommended actions, reducing false positives and mean time to respond (MTTR).
- **Continuous adaptation** — online baseline updates, cold-start handling, and behavioral-drift monitoring keep detection accurate as user behavior evolves.
- **Centralized operations** — a single SOC dashboard delivers live alerts, behavioral analytics, geospatial threat mapping, and interactive investigation for efficient response.

## 1.4 Innovation & Uniqueness

- **Hybrid two-stage AI pipeline** — combines label-free unsupervised anomaly detection (catches novel patterns) with supervised attack classification (names known patterns), improving accuracy while keeping compute low by only classifying events already flagged as suspicious.
- **16 behavioral-intelligence features** — models users across temporal, geographical, authentication, device, and resource-access dimensions, including impossible-travel velocity, device familiarity, and privilege-escalation signals — not static signatures.

- **Explainable AI at the point of decision** — SHAP converts model output into one-sentence justifications a SOC analyst can read in under two seconds, building analyst trust and auditability.
- **SOC-native enrichment layer** — every alert is auto-mapped to a MITRE ATT&CK technique and paired with a severity score and a ranked, transparent response playbook, so alerts speak the language every SOC already uses.
- **Production-ready by design** — explicitly engineers for cold-start users, concept drift, severe class imbalance, and real-time streaming, and ships an end-to-end platform with an AI Copilot, investigation inspector, and analyst-feedback loop.

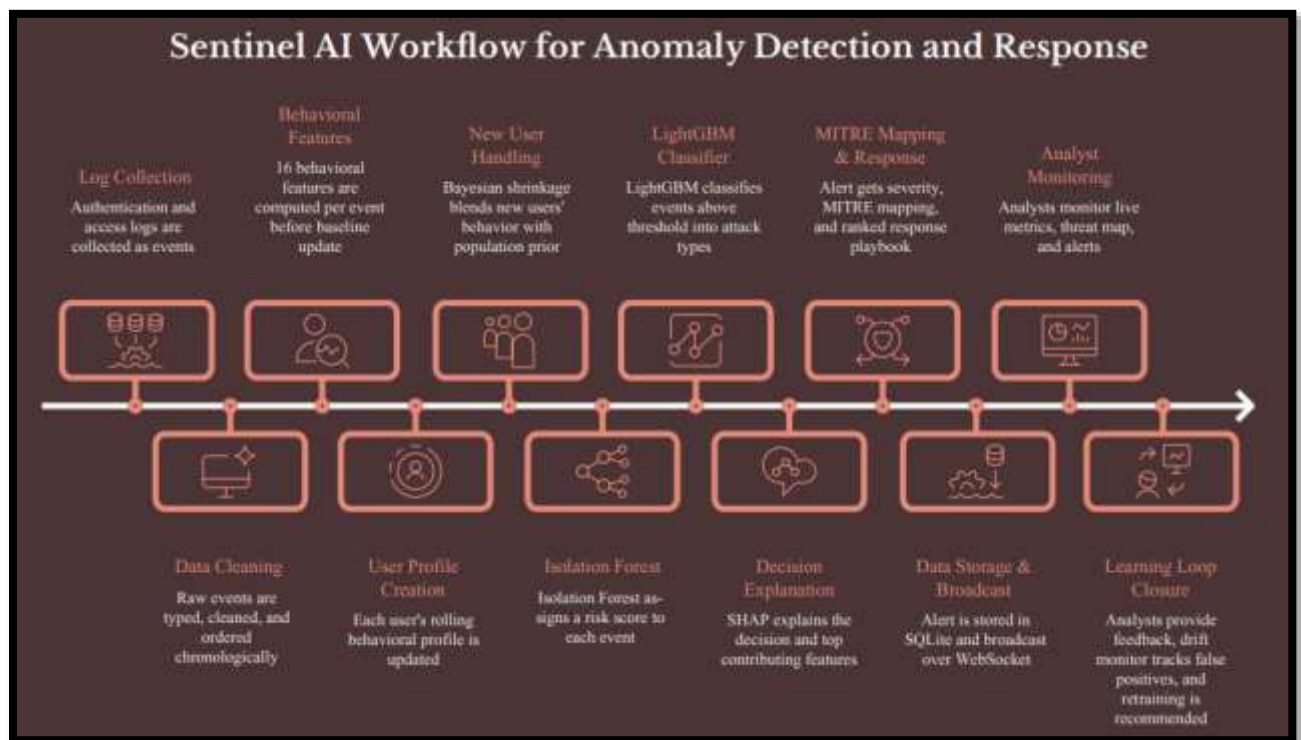


Figure 1 — Sentinel AI end-to-end architecture: ingestion → feature engineering → two-stage detection → enrichment → SOC dashboard.

## 2. System Capabilities & Feature Catalog

Sentinel AI is more than a detection model — it is a complete Security Operations platform. The table below summarizes the shipped capabilities that make it deployable in a real SOC.

| Capability                   | What it delivers                                                                                                                                                            |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UEBA Behavioral Baselining   | Learns a personalized behavioral profile per user (typical hours, home location, known devices, resource habits, session norms) and scores every event against it.          |
| Two-Stage Detection Engine   | Unsupervised anomaly scoring (Isolation Forest) followed by a supervised LightGBM attack-type classifier across five attack categories.                                     |
| Explainable AI (SHAP)        | Converts model output into a one-line, plain-English reason for every alert, with the top contributing behavioral features surfaced to the analyst.                         |
| MITRE ATT&CK Enrichment      | Automatically maps each detected attack to a real ATT&CK technique ID and tactic (e.g. T1110 Brute Force, T1078 Valid Accounts, T1021 Remote Services, T1036 Masquerading). |
| Severity Triage              | Blends anomaly risk with asset sensitivity into a Critical / High / Medium / Low severity, with escalation when high-impact techniques hit sensitive assets.                |
| Automated Response Playbooks | Attaches ranked, transparent recommended actions per alert (e.g. Block source IP, Require MFA, Isolate host, Page on-call analyst).                                         |
| Real-Time Alert Streaming    | WebSocket-based live alert feed pushes new detections to the SOC dashboard the instant they occur.                                                                          |
| AI Security Copilot          | Generates analyst briefings and answers questions about individual alerts, turning raw detections into investigation-ready narratives.                                      |
| Interactive Investigation    | A User Profile Inspector lets analysts drill into any user's baseline, recent events, and associated alerts on demand.                                                      |
| Geospatial Threat Map        | Network Attack Replay visualizes attack origins and arcs toward the enterprise HQ for at-a-glance situational awareness.                                                    |
| Cold-Start Handling          | Bayesian shrinkage blends population priors with sparse user history so new users are neither ignored nor over-flagged.                                                     |
| Drift Monitoring             | Tracks analyst feedback to detect model staleness (rising false-positive rate) and recommends retraining.                                                                   |
| Class-Imbalance Handling     | Inverse-frequency class weighting plus optional SMOTE oversampling so rare attack types are still learned.                                                                  |
| Analyst Feedback Loop        | Analysts confirm or dismiss alerts; verdicts feed drift monitoring and future retraining.                                                                                   |
| Admin / Analyst Access       | Gated analyst login controls access to the operations console.                                                                                                              |
| Live Replay Engine           | Replays historical events as a simulated live stream to demonstrate and stress-test the end-to-end pipeline.                                                                |
| Metrics & Health API         | Exposes live KPIs: alert rate, active users, average risk, detection accuracy, false-positive rate, AI confidence, and severity mix.                                        |

## 3. Technical Approach

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### 3.1 Data Sources

The prototype uses a realistic synthetic enterprise authentication dataset that simulates normal activity and multiple attack scenarios, while the architecture is designed to ingest real-world logs.

- **Authentication & access logs** — login events with timestamps, IP addresses, devices, geolocation, authentication status, accessed resources, resource sensitivity, and session details.
- **Behavioral profiles** — rolling per-user history used to build and update personalized baselines.
- **Attack injection engine** — generates labeled events for Credential Misuse, Brute Force, Lateral Movement, Impossible Travel, and Device Spoofing for training and evaluation.

### 3.2 Technologies & Tools

| Layer                | Stack                                                                            |
|----------------------|----------------------------------------------------------------------------------|
| Languages            | Python (backend & ML), JavaScript (frontend)                                     |
| Backend / API        | FastAPI, WebSocket, SQLAlchemy, Uvicorn                                          |
| Machine Learning     | scikit-learn (Isolation Forest), LightGBM, SHAP, imbalanced-learn (SMOTE)        |
| Data                 | Pandas, NumPy                                                                    |
| Frontend / Dashboard | HTML5, CSS3, JavaScript, Canvas-based charts (React-ready design)                |
| Storage              | SQLite (prototype); pluggable for enterprise databases                           |
| Deployment           | FastAPI service + SOC dashboard with live WebSocket streaming (hosted on Render) |

### 3.3 Data & Detection Pipeline

- Ingest authentication logs (synthetic or live) in time order.
- Engineer 16 behavioral features per event — computed *before* the event updates the baseline, so no event “sees itself.”
- Baseline each user with cold-start-aware profiles that blend population priors with individual history.
- Detect anomalies via the unsupervised scorer, producing a 0–1 risk score.
- Classify flagged anomalies into one of five attack types with LightGBM.
- Enrich each alert with severity, MITRE ATT&CK technique, and a ranked response playbook.
- Explain & stream the alert (SHAP sentence) live to the SOC dashboard over WebSocket.
- Learn from analyst feedback and monitor behavioral drift to trigger retraining.

### 3.4 Behavioral Feature Intelligence

The feature set captures the behavioral context that signatures miss, grouped into five dimensions:

- **Temporal** — hour-of-day and hour typicality (how unusual this login time is for the user), time since last event.

- **Geographical** — new-country flag, distance from home location, and impossible-travel velocity (implied km/h since the last login).
- **Authentication** — failed-auth flag and failed attempts in the last 5 minutes (brute-force signal).
- **Device** — new-device flag and device familiarity (how established the device is for this user).
- **Resource & session** — resource sensitivity and familiarity, session duration and its deviation from expectation, privilege-escalation flag, and a baseline-confidence signal for cold-start awareness.

### 3.5 Models & Methods

- **Behavioral profiling** — learns normal per-user behavior across the 16 engineered features.
- **Unsupervised anomaly detection** — Isolation Forest scores each event's abnormality with no labels, so it operates from day one and can flag novel patterns; the raw score is calibrated into a stable 0–1 risk range.
- **Supervised classification** — LightGBM (with a GradientBoosting fallback) categorizes flagged anomalies into the five attack types, trained with class weighting and SMOTE.
- **Explainable AI** — SHAP TreeExplainer produces per-alert feature attributions and a plain-English sentence; a z-score fallback explains anomalies even with no attack label.
- **Enrichment & response** — rule-based, auditable mapping to severity, MITRE ATT&CK, and recommended actions.
- **Drift monitoring** — rolling false-positive rate plus an ADWIN-style windowed comparison recommend retraining when the model goes stale.

### 3.6 SOC Dashboard & Analyst Experience

- **Live Operations view** — KPI strip (alerts, events, detection F1, active classes), severity strip, live alert stream, per-class recall, model-health/drift, risk-score trend, and severity mix.
- **Network Attack Replay** — geospatial map animating attack origins arcing toward the enterprise HQ.
- **Explainability panel** — shows the AI-generated reason and top contributing behavioral features behind each alert, with MITRE context and recommended actions.
- **Investigation inspector** — search any user to inspect their baseline, recent activity, and alerts.
- **AI Security Copilot** — one-click “Ask Copilot” turns any alert into an analyst briefing.



## 4. Feasibility & Viability

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### 4.1 Feasibility Analysis

- **Data availability** — validated on a realistic synthetic dataset with labeled attacks; the architecture is designed to integrate real logs from identity providers, VPNs, cloud platforms, and SIEM/SOAR tools.
- **Technical feasibility** — built on widely adopted, well-supported technologies (FastAPI, scikit-learn, LightGBM, SHAP, SQLite), making it practical to develop, deploy, and scale.
- **Deployment feasibility** — a modular architecture with REST APIs, WebSocket streaming, and database integration supports seamless deployment into modern SOC and SIEM environments.

### 4.2 Challenges & Risks

- **Class imbalance** — real logs contain very few attacks relative to legitimate activity, making naive training collapse to “predict normal.”
- **Behavioral drift** — user behavior shifts over time (new devices, roles, locations), degrading accuracy if unaddressed.
- **Cold-start users** — new users lack the history needed for a reliable baseline.
- **Scalability** — high event volumes demand efficient, low-latency feature engineering and scoring.

### 4.3 Mitigation Strategies

- **Imbalance handling** — inverse-frequency class weighting plus SMOTE oversampling on the training split so rare attack classes are learned.
- **Adaptive learning** — continuous baseline updates and drift monitoring maintain detection performance over time.
- **Cold-start handling** — Bayesian shrinkage blends organizational priors with limited user history until enough behavior accumulates.
- **Efficient architecture** — the two-stage design classifies only already-flagged events, and incremental baseline updates minimize overhead for real-time detection.

## 5. Performance Evaluation & Results

### 5.1 Experimental Setup

The system was evaluated on a synthetic enterprise authentication dataset of **20,351 events** — **19,200 normal** and **1,151 labeled attacks** across five categories. Evaluation ran in three stages: (i) behavioral anomaly detection, (ii) attack classification, and (iii) an end-to-end replay simulation of real-time performance.

### 5.2 Stage 1 — Behavioral Anomaly Detection

The unsupervised anomaly scorer was evaluated with standard anomaly-detection metrics.

| Metric    | Value  |
|-----------|--------|
| ROC-AUC   | 0.9855 |
| Precision | 87.1%  |
| Recall    | 76.3%  |
| F1-Score  | 81.3%  |

Confusion matrix at the selected operating threshold:

| True Neg. | False Pos. | False Neg. | True Pos. |
|-----------|------------|------------|-----------|
| 19,070    | 130        | 273        | 878       |

These results demonstrate excellent separation between normal and malicious behavior while maintaining a low false-positive rate (130 of 19,200 normal events).

### 5.3 Stage 2 — Attack Classification

Events flagged in Stage 1 were classified with LightGBM.

| Metric          | Value |
|-----------------|-------|
| Macro Precision | 100%  |
| Macro Recall    | 100%  |
| Macro F1-Score  | 100%  |

The classifier correctly identified all five attack categories on the held-out synthetic test set, indicating strong discrimination between distinct attack behaviors.

### 5.4 End-to-End Replay Evaluation

To test the complete pipeline, 4,071 authentication events were replayed as a simulated live stream.

| Metric                      | Value        |
|-----------------------------|--------------|
| Live events replayed        | <b>4,071</b> |
| True attacks                | <b>231</b>   |
| Attacks detected            | <b>166</b>   |
| End-to-end detection recall | <b>71.9%</b> |

Attack-wise detection performance:

| Attack Type       | Detected / Total | Rate         |
|-------------------|------------------|--------------|
| Brute Force       | 128 / 128        | <b>100%</b>  |
| Credential Misuse | 9 / 9            | <b>100%</b>  |
| Device Spoofing   | 6 / 6            | <b>100%</b>  |
| Impossible Travel | 8 / 16           | <b>50%</b>   |
| Lateral Movement  | 15 / 72          | <b>20.8%</b> |

The pipeline reliably catches high-confidence attack patterns (brute force, credential misuse, device spoofing) with a low false-positive rate. Subtle, low-and-slow behaviors such as lateral movement are harder to detect and represent the primary direction for future enhancement.

## 5.5 MITRE ATT&CK Coverage

Each detected attack type is automatically mapped to a recognized ATT&CK technique, so alerts integrate directly into existing SOC workflows.

| Attack Type       | Technique | Tactic               |
|-------------------|-----------|----------------------|
| Brute Force       | T1110     | Credential Access    |
| Credential Misuse | T1078     | Privilege Escalation |
| Lateral Movement  | T1021     | Lateral Movement     |
| Impossible Travel | T1078     | Initial Access       |
| Device Spoofing   | T1036     | Defense Evasion      |

## 5.6 Key Outcomes

- Generated and evaluated 20,351 enterprise authentication events with injected attack scenarios.
- Achieved a ROC-AUC of 0.9855 with 87.1% precision, 76.3% recall, and 81.3% F1 for behavioral anomaly detection.
- Reached 100% macro precision, recall, and F1 for attack-type classification on the held-out set.
- Detected and classified 166 cyberattacks during live replay at 71.9% end-to-end recall, with 100% detection on brute force, credential misuse, and device spoofing.

- Demonstrated real-time detection, explainable risk scoring, MITRE-enriched alerts, and geospatial threat visualization through the SOC dashboard, Copilot, and investigation inspector.

**Note on results** — *The reported metrics are obtained on a synthetically generated enterprise dataset with injected attacks. They validate the correctness and effectiveness of the architecture but should not be read as production-level performance on real-world enterprise traffic.*

## 6. Future Enhancements

- **Deeper lateral-movement detection** — graph-based session and host-to-host modeling to catch low-and-slow movement missed by per-event scoring.
- **Deep-learning anomaly models** — swap in the stubbed autoencoder / sequence models for richer temporal behavior modeling.
- **Live enterprise integrations** — connectors for identity providers, VPNs, and SIEM/SOAR to move from synthetic to production telemetry.
- **Automated response (SOAR)** — promote recommended playbooks into one-click or automated containment actions.
- **Scale-out storage & streaming** — move from SQLite to a distributed store and event bus for enterprise event volumes.

## 7. Research & References

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- **MITRE ATT&CK Framework** — Enterprise knowledge base of adversary tactics and techniques used to model attack behaviors (credential access, lateral movement, privilege escalation).  
<https://attack.mitre.org/>
- **NIST Cybersecurity Framework (CSF) 2.0** — Industry-standard framework for cybersecurity risk management, detection, response, and recovery.  
<https://www.nist.gov/cyberframework>
- **scikit-learn: Machine Learning in Python** — Core ML library used for anomaly detection and preprocessing.  
<https://scikit-learn.org/stable/>
- **LightGBM Documentation** — Gradient-boosting framework used for supervised attack classification.  
<https://lightgbm.readthedocs.io/>
- **SHAP (SHapley Additive exPlanations)** — Explainable-AI framework used to interpret predictions and generate analyst-friendly explanations.  
<https://shap.readthedocs.io/>
- **Pedregosa, F., et al. (2011)** — “Scikit-learn: Machine Learning in Python.” Journal of Machine Learning Research, 12, 2825–2830.  
<https://jmlr.org/papers/v12/pedregosa11a.html>

## 8. Prototype — Live Demo & Repository

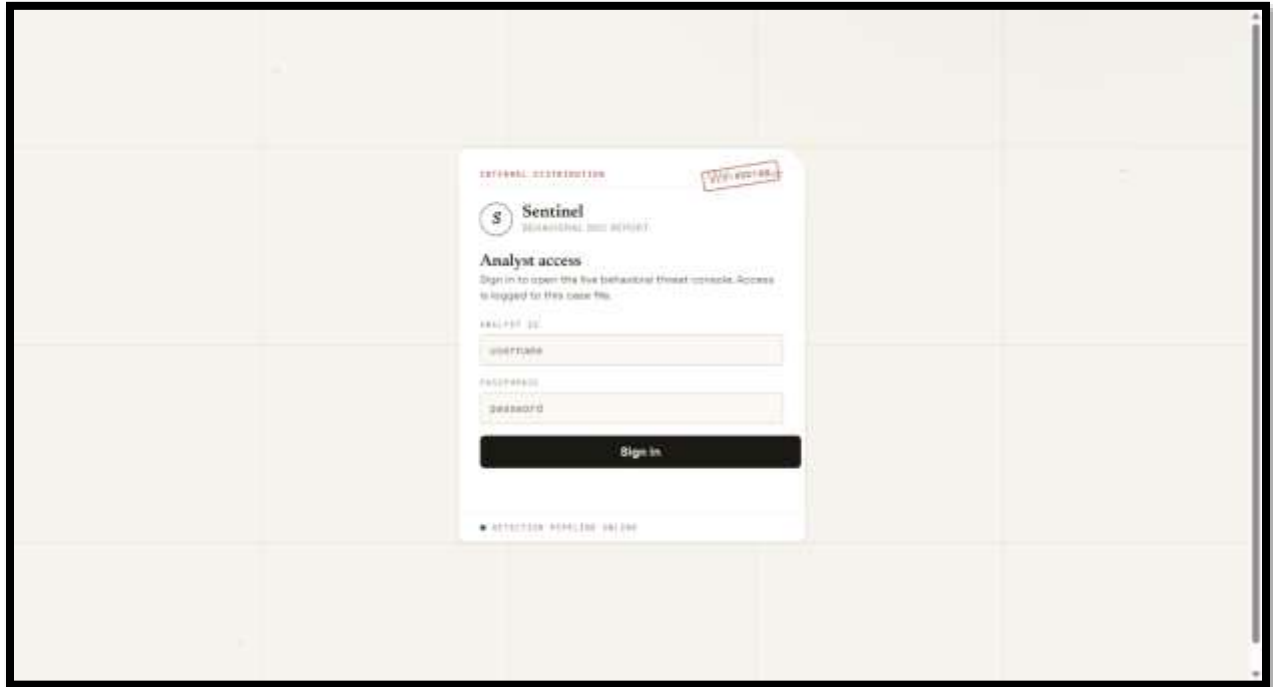
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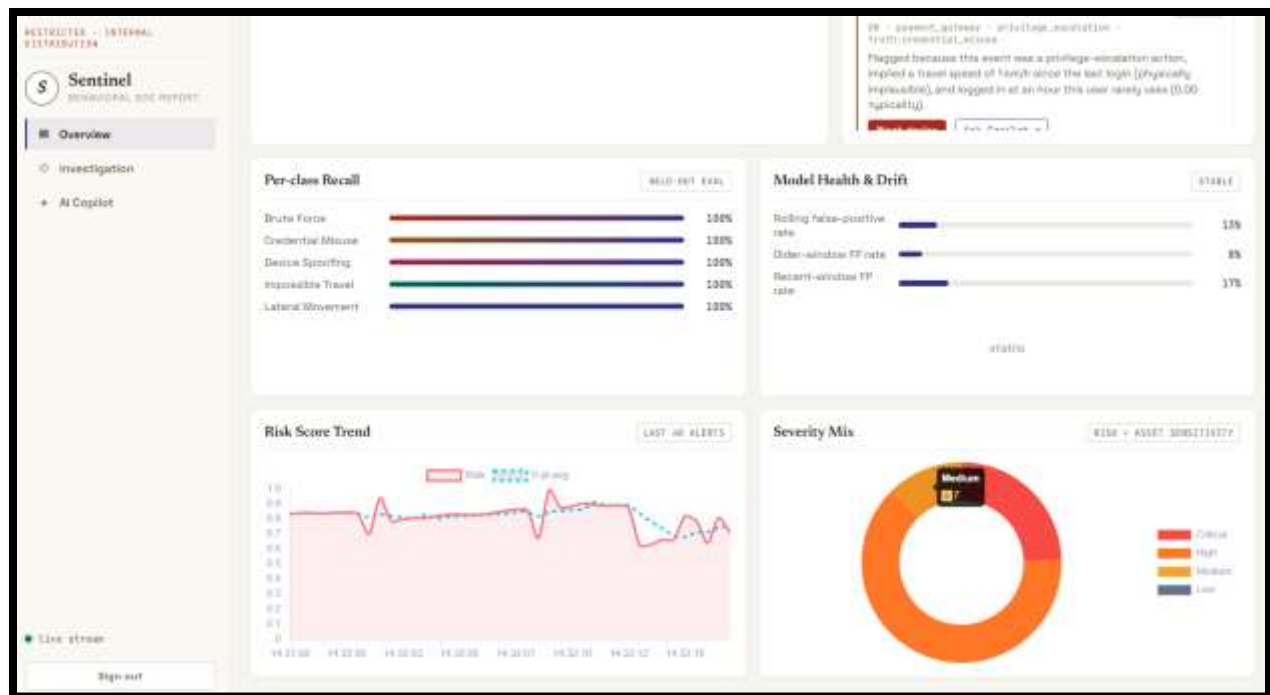
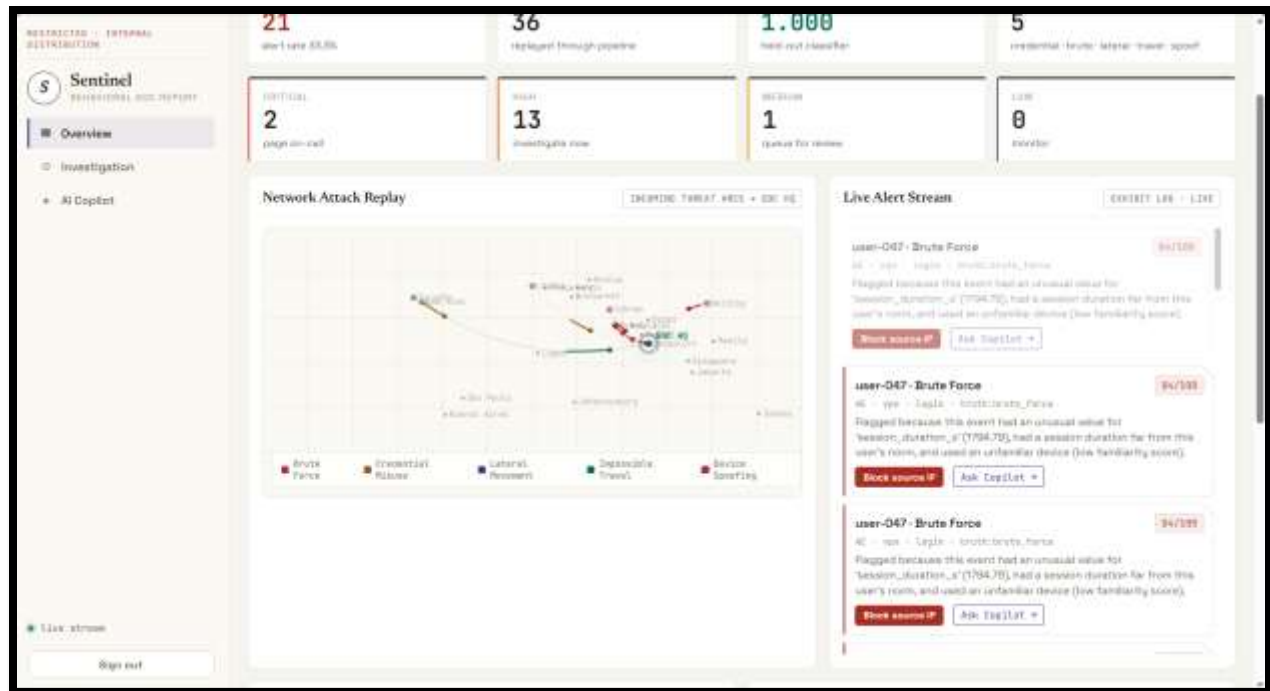
*Note: data shown on the live site uses assumed/sample values at the prototype stage. Once the model is fully productionized, computed results will be integrated into the site.*

**Live Website:** [Link](#) OR <https://sentinels-soc.onrender.com/>

**GitHub Repository:** [GitHub](#) OR <https://github.com/AaryaButolia11/Sentinels-SOC>

**Screenshots:**





RESTRICTED - INTERNAL  
RESTRICTION

Sentinel

SECURITY ANALYTICS

Overview

Investigation

AI Copilot

Live stream

Sign out

INTERACTIVE INVESTIGATION

User Profile Inspector

Inspect the learned baseline the detector scores each event against

Investigate

user-031

user-047 - 254ev - 48X   user-011 - 255ev - 378   user-057 - 344ev - 368   user-007 - 239ev - 368   user-026 - 239ev - 528   user-006 - 122ev - 27X

user-018 - 135ev - 26X   user-043 - 233ev - 258   user-071 - 328ev - 218   user-033 - 238ev - 218   user-084 - 234ev - 218   user-015 - 227ev - 28X

user-031

0 alerts on record

HYDRA OBSERVED

255

COMMON SOURCE

dev-031a231a,  
dev-0a7d784c,  
dev-1a9d88e7,  
dev-2874b99b,  
dev-22757a45,  
dev-2a8a5c22,  
dev-2a85d483,  
dev-34ba611f,  
dev-4b95895e,  
dev-415d2a88,  
dev-aff5a0ef,  
dev-a257c0aa,  
dev-af9d3b7b,  
dev-71a7bdf6,  
dev-81a20fab,  
dev-831d29d5,  
dev-8e9c0511,  
dev-92a950c8,  
dev-9a8a8314,  
dev-94ce33ae,  
dev-a5d00a41,  
dev-a62f11ab,  
dev-a571ac5a,  
dev-b75e30f4,  
dev-bfb6b79e,  
dev-ca78f68e.

COMMON COUNTRY

CA, IN, MD, PH,  
UA

PHYSICAL HOSTS

T, R, 10, 16, 10,  
20

SCORE (1.0 = 100)

41.1845, -63.196

Avg SCORES

1709.7s

PERIOD AUTH

39

Live stream

Sign out

RESTRICTED - INTERNAL  
RESTRICTION

Sentinel

SECURITY ANALYTICS

Overview

Investigation

AI Copilot

Live stream

Sign out

dev-031a231a,  
dev-a91d15c7,  
dev-ec1fa1c3,  
dev-e6115e58,  
dev-a773a689,  
dev-f8a14d6e,  
dev-f8d734f5,  
dev-f8d574ab,  
dev-f8e13a75,  
dev-fc7e4fa2

TOP RESOURCES

shared\_drive,  
source\_repo, vpn,  
admin\_console,  
finance\_share

Baseline confidence

ESTABLISHED

97.7%

Risk timeline

9 POINTS

No risk finding yet.

Recent events

LAST 5

| TIME        | RESOURCE    | ACTION      | COUNTRY | AUTH    |
|-------------|-------------|-------------|---------|---------|
| 07-26 07:32 | source_repo | file_access | CA      | SUCCESS |
| 07-25 10:37 | source_repo | read        | CA      | SUCCESS |
| 07-25 10:05 | source_repo | file_access | CA      | SUCCESS |

Alert history

0

No alerts on record for this user.

RESTRICTED - INTERNAL DISTRIBUTION

S

Sentinel

ANALYTICAL SOC REPORT

Overview

Investigation

AI Copilot

Live stream

Sign out

41 SECURITY COPILOT

Analyst Briefings

Grounded in each alert's own evidence - MITRE ATT&CK mapping & response playbook

1,200 - 8,000 - 11,000 - 15,000 - 20,000 - 25,000

Recent alerts

28

Credential Misuse

user-076 - 14:12:13

71

Device Spoofing

user-002 - 14:12:14

81

Credential Misuse

user-002 - 14:12:14

64

Device Spoofing

user-002 - 14:12:15

79

Impossible Travel

user-034 - 14:12:16

82

Lateral Movement

user-048 - 14:12:16

66

Lateral Movement

user-048 - 14:12:16

66

Lateral Movement

user-002 - 14:12:17

63

Lateral Movement

user-002 - 14:12:17

63

Impossible Travel

87

Device Spoofing

8184

118

14:12:14/16

MITRE ATT&CK - T1036 (Masquerading) / T1050 (Use of Alternative Auth Material)

\*\*Situation Summary\*\*

A high-risk alert has been raised for a potential device spoofing attack on the admin console, with a user from an unfamiliar country attempting to log in. The model flagged this event due to the high sensitivity of the resource, unusual country of origin, and anomalous session duration.

\*\*MITRE ATT&CK Technique\*\*

T1027 (Obfuscated Files or Information) is not directly applicable, however T1114 (Email Collection) is also not relevant. The closest match is T1078 (Valid Accounts), as the attack involves a legitimate user account.

\*\*Containment/Response Actions\*\*

1. Verify user-002's identity and recent travel history.

2. Review admin console access logs for similar anomalies.

3. Temporarily restrict user-002's access to the admin console.

\*\*Threat-Hunting Pivot\*\*

Investigate other recent logins from unfamiliar countries to identify potential reconnaissance or lateral movement attempts.

RECOMMENDED RESPONSE

1. De-trust the device identifier and require full re-enrollment.

2. Invalidate any device-bound tokens or certificates tied to that ID.

3. Verify with the owner whether the real device was lost, stolen, or cloned.

4. Tighten device-attestation / conditional-access policy for the affected resource.

Threat hunt: Look for the genuine device active elsewhere at the same time -- concurrent use confirms the spoof.

RESTRICTED - INTERNAL DISTRIBUTION

S

Sentinel

ANALYTICAL SOC REPORT

Overview

Investigation

AI Copilot

Live stream

Sign out

41 SECURITY COPILOT

Analyst Briefings

Grounded in each alert's own evidence - MITRE ATT&CK mapping & response playbook

1,200 - 8,000 - 11,000 - 15,000 - 20,000 - 25,000

Recent alerts

29

Credential Misuse

user-076 - 14:12:17

71

Device Spoofing

user-002 - 14:12:18

81

Credential Misuse

user-002 - 14:12:18

64

Device Spoofing

user-002 - 14:12:18

79

Impossible Travel

user-034 - 14:12:19

82

Lateral Movement

user-048 - 14:12:19

66

Lateral Movement

user-048 - 14:12:19

66

Lateral Movement

user-048 - 14:12:20

63

Lateral Movement

user-048 - 14:12:20

63

Impossible Travel

87

Impossible Travel

8184

118

14:12:19/20

MITRE ATT&CK - T1078 (Valid Accounts) / geolocation anomaly

\*\*Situation Summary\*\*

An impossible travel alert has been raised for user-034, indicating a potential security incident due to a login from a device never seen before, with an unusually short session duration and an implausibly high travel speed since the last login. The risk score of 82/100 suggests a high likelihood of malicious activity.

\*\*MITRE ATT&CK Technique\*\*

T1078 (Valid Accounts)

\*\*Containment/Response Actions\*\*

1. Immediately suspend user-034's account to prevent further potential malicious activity.

2. Investigate the device used for the suspicious login and its connection to the VPN.

3. Review recent login history for user-034 to identify any other potential security incidents.

\*\*Threat-Hunting Pivot\*\*

Investigate other users who have logged in from the same origin country (CA) in the last 24 hours to identify potential related activity.

RECOMMENDED RESPONSE

1. Treat the second (anomalous) session as hostile and terminate it.

2. Force a global credential reset and MFA re-enrollment for the account.

3. Confirm the user's true location out-of-band; rule out VPN/proxy false positives before escalating.

4. Review actions taken during the anomalous session for data access or config changes.

Threat hunt: Check whether the far-side IP appears for other accounts -- shared malicious infrastructure is common.



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Analyst Briefings

Grounded in each alert's own evidence - MITRE ATT&CK mapping & response playbook

117K · 100% · 2,400-5,000-10,000-100,000+

Recent alerts

23

Credential Misuse

user-074 · 14:22:17

71

Device Spoofing

user-003 · 14:22:16

81

Credential Misuse

user-003 · 14:22:16

64

Device Spoofing

user-025 · 14:22:15

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Impossible Travel

user-034 · 14:22:15

82

Lateral Movement

user-040 · 14:22:14

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Lateral Movement

user-040 · 14:22:14

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Lateral Movement

user-040 · 14:22:13

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Lateral Movement

user-040 · 14:22:13

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Impossible Travel

user-040 · 14:22:13

87

Credential Misuse

HIGH

117K

100%

2,400-5,000-10,000-100,000+

MITRE ATT&CK - T1078 (Valid Accounts)

"Situation Summary": A potential credential misuse attack has been detected on the payment gateway resource, with a risk score of 71/100, targeting the user-074 account. The alert was triggered by a privilege escalation action that appears to be physically implausible, given the short time frame and zero travel speed since the last login.

"MITRE ATT&CK Technique": T1078: Valid Accounts (Privilege Escalation)

"Containment/Response Actions":

1. Immediately disable the user-074 account.

2. Initiate a password reset for the affected account.

3. Review recent activity on the payment gateway for suspicious transactions.

"Threat Hunting Pivot": Investigate other accounts with similar login patterns and short time frames between actions to identify potential lateral movement.

RECOMMENDED RESPONSE

1. Contact the account owner out-of-band to confirm the access was intentional.

2. Review what was read/exfiltrated from the sensitive resource during the session.

3. Suspend the session and require re-authentication with MFA.

4. Check for privilege changes made in the same window.

Threat hunt: Compare this access against the user's 30-day resource baseline; look for a preceding phishing or token-theft event.

Thank You

Sentinel AI · Behavioral SOC · Aarya Butolia